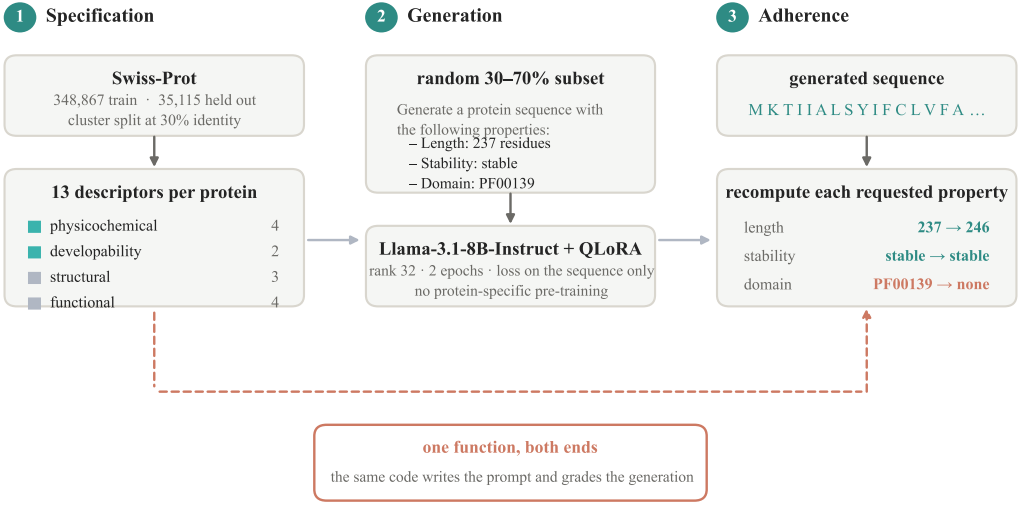
1
Specification
Swiss-Prot
348,867 train · 35,115 held out
cluster split at 30% identity
13 descriptors per protein
physicochemical 4
developability 2
structural 3
functional 4
2
Generation
random 30–70% subset
Generate a protein sequence with
the following properties:
– Length: 237 residues
– Stability: stable
– Domain: PF00139
Llama-3.1-8B-Instruct + QLoRA
rank 32 · 2 epochs · loss on the sequence only
no protein-specific pre-training
3
Adherence
generated sequence
M K T I I A L S Y I F C L V F A …
recompute each requested property
length 237 → 246
stability stable → stable
domain PF00139 → none
one function, both ends
the same code writes the prompt and grades the generation